

This document describes the data instances used in the computational experiments of the paper:

“A two-stage machine learning-assisted branch-and-price framework for integrated home healthcare shift scheduling and routing.”

Instance Overview

The problem integrates caregiver selection, shift planning, patient assignment, visit scheduling, and routing. The instances also include pair-specific care intensity values used to represent differences in caregiver workload.

The benchmark contains 54 instances covering six patient-set sizes and three spatial structures. Three independent replications are provided for each combination of patient-set size and spatial structure.

The instances are classified into three size groups:

Instance class	Number of patients	Candidate caregivers	Number of instances
Small	15	5-6	9
	25	7-9	9
Medium	40	10-12	9
	60	12-15	9
Large	80	15-18	9
	100	18-20	9
Total	-	-	54

For each problem size, the benchmark includes three spatial structures:

- `C`: clustered patient locations
- `R`: randomly distributed patient locations
- `RC`: mixed random-clustered patient locations

Three independent replications are generated for each size-structure combination.

File-naming convention

Instance files follow this convention:

`case\_<replication>\_<patients>\_<caregivers>\_<type>.txt`

For example:

`case\_1\_15\_5\_C.txt`

denotes:

- replication: 1

- number of patients: 15
- number of candidate caregivers: 5
- spatial structure: clustered (`C`)

Parameter Generation Scheme

- **Node indexing**

Each instance contains two depot copies in addition to the patient nodes.

For an instance with $|P|$ patients:

- node `0` is the starting depot;
- nodes $1, \dots, |P|$ are patient nodes;
- node $|P|+1$ is the ending depot.

The starting and ending depot nodes represent the same physical healthcare center and therefore have the same spatial coordinates. Consequently, the total number of nodes is:

$$\text{number_of_nodes} = \text{number_of_patients} + 2.$$

- **Planning horizon**

The planning horizon consists of seven days and fourteen shifts:

- morning shift: 08:00--13:00;
- afternoon shift: 13:00--18:00.

Each patient requires one service during the planning horizon. Depending on the instance, each service can be performed in two to four alternative shifts on distinct days.

- **Spatial generation**

Patient locations are generated in the two-dimensional region:

$[0,100] \times [0,100]$

The healthcare center is located at:

$(50,50)$

The three spatial structures are generated as follows:

- `R`: patient coordinates are randomly distributed over the service region;
- `C`: patient coordinates are generated around two to four cluster centers;
- `RC`: 60% of patient locations are clustered, and 40% are randomly distributed.

The number of cluster centers depends on the instance size.

- **Travel times**

Travel times are deterministic and are calculated from Euclidean distances using an average travel speed of 55 km/h. The values stored in the instance files are expressed in minutes. The travel-time matrix includes both depot copies and all patient nodes. Its dimensions are:

`number\_of\_nodes x number\_of\_nodes`

- **Service times**

Patient service durations are generated from the discrete uniform distribution:

`U{20,...,60}` minutes.

The service duration of each depot node is zero.

- **Patient time windows**

Patients are assigned to one of three time-window classes:

- tight: probability 0.30 and width 30 minutes;
- medium: probability 0.50 and width between 60 and 120 minutes;
- wide: probability 0.20 and width equal to the entire shift.

For each shift, two vectors are reported:

- `alpha`: earliest service-start times;
- `beta`: latest service-start times.

A zero value for both `alpha` and `beta` indicates that the corresponding patient is not available in that day.

- **Caregiver categories**

Candidate caregivers may belong to one of three categories:

Category	Probability	Activation-cost range
Junior	0.35	100--130
Senior	0.40	131--170
Expert	0.25	171--220

The generation procedure ensures that each patient is compatible with at least:

- 3 caregivers in small instances;
- 4 caregivers in medium instances;
- 5 caregivers in large instances.

- **Skill compatibility**

The caregiver-patient compatibility information identifies the patients that can be served by each caregiver.

If patient `i` is listed for caregiver `k`, the caregiver is qualified to serve that patient. Otherwise, the assignment is incompatible.

- **Care intensity**

The Care Intensity Instrument value is defined for each compatible caregiver-patient pair. The following ranges are used:

- junior caregiver: `[6,10]`;
- senior caregiver: `[3,7]`;
- expert caregiver: `[1,5]`.

Lower values indicate better caregiver-patient suitability. A value of zero denotes an incompatible caregiver-patient assignment.

The care-intensity matrix has dimensions:

`number\_of\_caregivers x number\_of\_nodes`

Instance-file structure

Each structured instance file contains the following sections:

1. instance metadata;
2. caregiver activation costs;
3. caregiver categories;
4. caregiver-patient skill;
5. service durations;
6. patient time windows;
7. caregiver-patient care-intensity matrix;
8. node-to-node travel-time matrix.